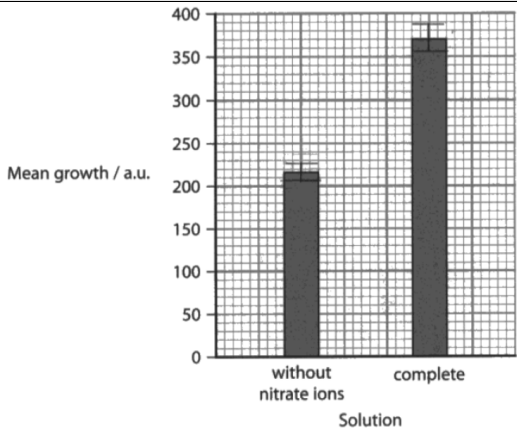


Question Number	Answer	Additional Guidance	Mark
1(a)(i)	An answer that includes the following points: • nitrate / sulfate / ammonium (1) • magnesium / nitrate (1) • calcium (1)	DO NOT ACCEPT formulae / ammonia DO NOT ACCEPT more than one if one is incorrect	(3)

Question Number	Answer	Additional Guidance	Mark
1(a)(ii)	An answer that includes four of the following points: • control of {species / type / age} of plant (1) • control of temperature / soil pH / light intensity / time of growth (1) • complete solution described and solution without named ion (1) • measurement of dependent variable described (1) • replication to measure variability / find SD (1)	ACCEPT if named ion in a list of correct ions Eg height / mass / number / size of leaves	(4)

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	An answer showing the following steps: • difference in {means} calculated / both mean rates calculated (1) • rate calculated with appropriate units (1)	Correct answer with no working shown gains 2 marks e.g. $371 - 216$ or 155 OR $371 \div 21$ AND $216 \div 21$ e.g. $\div 21 = 7.38 / 7.4 \text{ a.u.}$ $17.67 - 10.29 = 7.38 / 7.4 \text{ au}$	(2)

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	An answer that includes the following: • SDs added correctly to both bars (1) • the SDs do not overlap which shows that the difference (between the means) is significant (1) • the results for the complete solution are {more variable / less reliable} (1)	 DO NOT ACCEPT the results are significant ACCEPT range bars / error bars ACCEPT converse	(3)